

dongwookwon

ai engineer • agentic ai systems

contact

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Seongnam-si, Republic
of Korea

dongwooky.github.io
github / dongwooky
in / dongwooky

languages

korean (native)
english (fluent)

skills

- Python, C/C++
- PyTorch, Transformers
- LangChain, LLMs
- Linux, Docker, CUDA
- GNNs, time-series
- biomedical signals

interests

multi-agent LLMs
anomaly detection
nuclear cybersecurity
amateur radio

education

- 2024–2026 **M.S.** in Computer Engineering, Kwangwoon University, Seoul
GPA 4.33 / 4.5 (98.1%). Bio-Computing & Machine Learning Laboratory.
- 2025 **Graduate Research Program**, University of Toronto
GPA 3.68 / 4.0. Mechanical and Industrial Engineering.
- 2019–2024 **B.S.** in Electronics and Communications Engineering, Kwangwoon University, Seoul
GPA 3.41 / 4.5 (88.1%).

experience

- 2025–Now **SURESOFT TECHNOLOGIES**, Seongnam, Korea
AI Engineer, Intelligent Agent Technology Team. Developing AI validation system using agentic AI architectures for enterprise model verification. Designing autonomous and collaborative agents for systematic AI verification.
- 2021–2025 **BIO-COMPUTING & ML LAB, KWANGWOON U.**, Seoul, Korea
Research Assistant. Deep-learning anomaly detection in time-series (ECG, cyber-physical systems). EMNLP 2025 (Oral), ACL 2026; two granted KR patents.
- 2025 **LG ELECTRONICS CANADA**, Toronto, ON, Canada
Visiting Researcher. Built LLM-driven assessment framework for evaluating agentic AI systems.
- 2022 **QUALCOMM INSTITUTE, UC SAN DIEGO**, San Diego, CA, USA
AI Development Project Participant, Summer Program. Personality-based drug-addiction prediction. Achievement Award.

publications

- 2026 **CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades** , *ACL 2026 – Industry Track (Poster)*
R. Chang[†], **D. Kwon[†]**, J. Lee[†], and N. Verma. [†]Equal contribution.
- 2025 **GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems**, *EMNLP 2025 – Industry Track (Oral)*
J. Lee[†], R. Chang[†], **D. Kwon[†]**, H. Singh, and N. Verma. [†]Equal contribution.
- 2024 **Hybrid GAT + Transformer for implantable-device anomaly detection** , *IBEC 2024 – Best Poster Silver*
D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park.
- 2023 **Enhancing medical device security with GNN-GRU anomaly detection model** , *KOSOMBE 2023 – Outstanding Paper Award*
D. Kwon, Y. Kang, and C. Park.
- 2025 **Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation** , *Data Mining & Knowledge Discovery (under review)*
D. Kwon et al.

awards

- 2022 **Achievement Award** , *Qualcomm Institute AI Development Project – UC San Diego*
Honored for project work during the summer program.
- 2021 **Gold Award (Ministerial)** , *2021 Smart Maritime Logistics Project – Minister of Oceans and Fisheries*
Top award for project on container detection and stack-layer identification.
- 2025 **Silver Award** , *23rd Embedded Software Competition (KIISE)*
Recognition at one of Korea's flagship embedded-software competitions.

patents

- 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time** , *KR 10-2961984*
Issued patent, Republic of Korea. April 30, 2026.
- 2025 **GPT API-Based Network Anomaly Detection** , *KR 10-2848814*
Issued patent, Republic of Korea. August 18, 2025.